Multi-Class Chest X-Ray Pathology Classification

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***Abstract*—Chest X-ray imaging is widely used for diagnosing thoracic diseases. In this work, we develop a deep learning framework for multi-class chest X-ray pathology classification. The proposed approach combines contrastive self-supervised learning with supervised fine-tuning. An EfficientNet-v2 backbone is first pretrained using the SimCLR contrastive learning framework to learn robust visual representations. The network is then fine-tuned using focal loss to address severe class imbalance in medical datasets. Threshold optimization and test-time augmentation are further applied to improve prediction reliability. Experimental results demonstrate that contrastive pretraining improves representation quality and enhances downstream classification performance.**

***Index Terms*—Chest X-ray, medical imaging, contrastive learning, SimCLR, EfficientNet, focal loss**

I. Introduction

Chest X-ray imaging is one of the most widely used diagnostic tools for detecting thoracic diseases such as pneumonia, cardiomegaly, and pleural effusion. Accurate interpretation of chest X-rays requires extensive clinical expertise and is often time-consuming. Consequently, automated medical image analysis systems have become an important research area in computer vision and healthcare.

Deep convolutional neural networks have achieved strong performance in medical imaging tasks. However, several challenges remain when applying deep learning to chest X-ray classification, including limited labeled data, severe class imbalance, and subtle visual differences between disease categories.

Self-supervised learning has recently emerged as a powerful paradigm for learning useful visual representations without requiring large amounts of labeled data. Contrastive learning methods such as SimCLR have demonstrated remarkable success in learning transferable representations.

In this work, we propose a two-stage training pipeline that combines contrastive representation learning with supervised fine-tuning. The model first learns general visual representations using contrastive learning and is then adapted for pathology classification using supervised training with focal loss.

II. Methodology

Our proposed training pipeline consists of two stages: self-supervised contrastive pretraining and supervised fine-tuning. An overview of the pipeline is illustrated in Fig. 2.

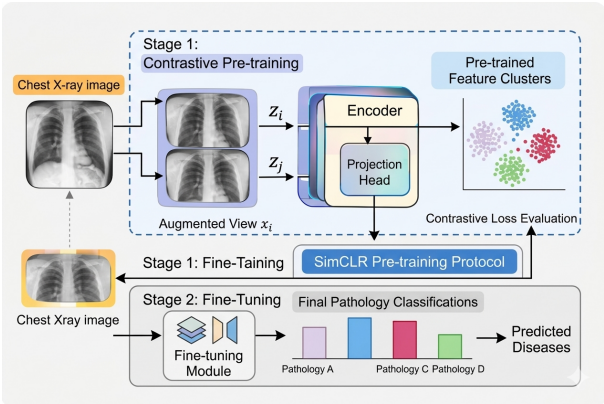

Fig. 1. Overview of the proposed chest X-ray classification pipeline. The model first learns visual representations using contrastive pre-training (SimCLR) and is subsequently fine-tuned for pathology classification.

A. Backbone Network

We employ EfficientNet-v2 as the backbone architecture due to its strong performance and efficient parameter scaling [1]. EfficientNet introduces a compound scaling method that balances network depth, width, and resolution to achieve improved accuracy with fewer parameters.

The backbone is initialized using ImageNet pretrained weights [4], which provides a strong initialization for feature extraction.

B. Contrastive Self-Supervised Pretraining

To improve representation learning, we perform contrastive pretraining using the SimCLR framework [2]. For each input image x, two stochastic augmentations are applied to generate two correlated views:

$$x_i = t_i(x), \quad x_j = t_j(x)$$

where t_i and t_j represent random image augmentations such as cropping, flipping, color jittering, and Gaussian blur.

Both views are passed through a shared encoder network $f(\cdot)$ followed by a projection head $g(\cdot)$ to obtain latent representations:

$$z_i = g(f(x_i)), \quad z_j = g(f(x_j))$$

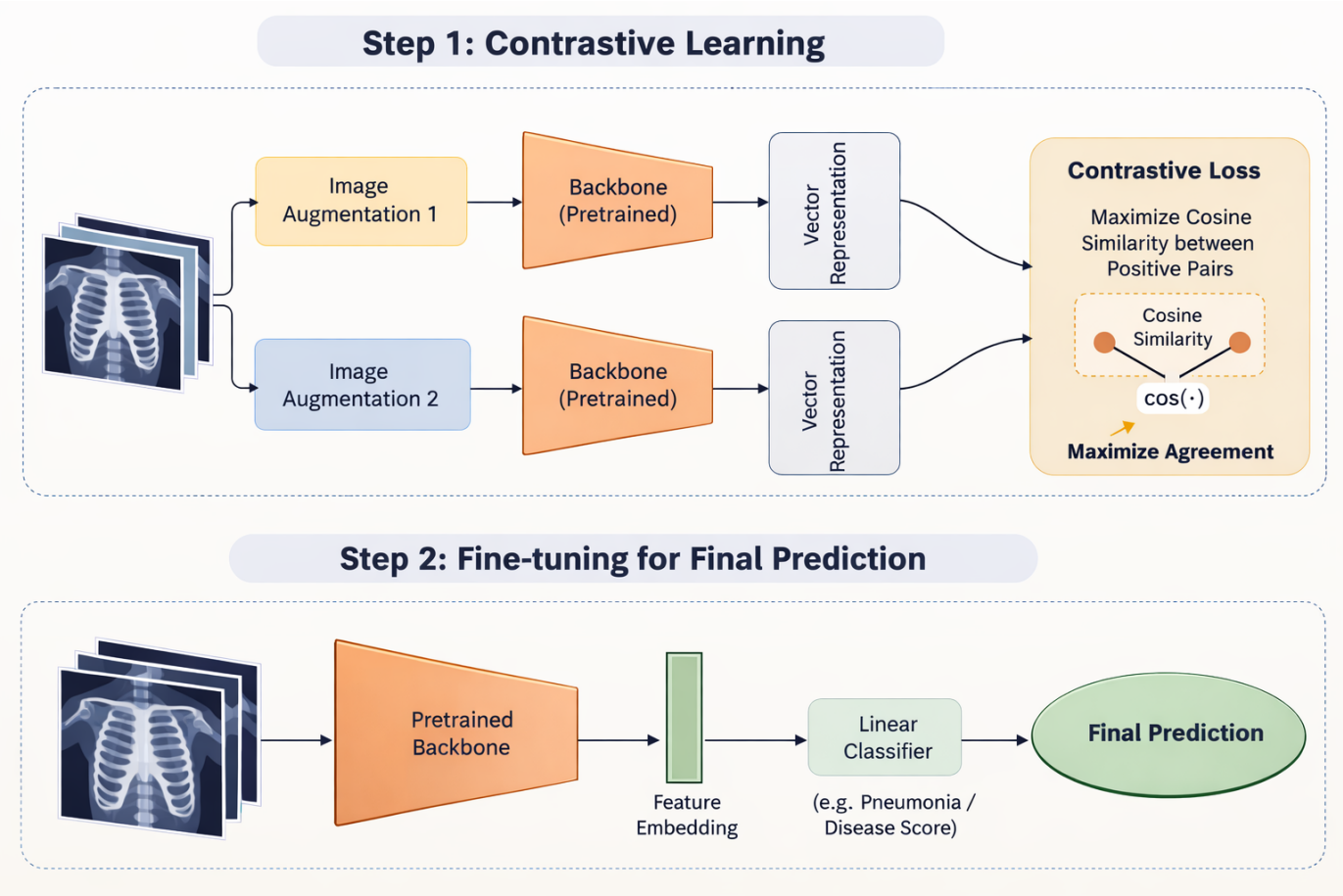

Fig. 2. Overview of the proposed training pipeline. In Stage 1, two augmented views of the same chest X-ray image are passed through a shared backbone network to obtain feature representations. A contrastive loss maximizes agreement between positive pairs. In Stage 2, the pretrained backbone is fine-tuned for multi-class pathology classification using a linear classifier.

The model is trained using the normalized temperature-scaled cross-entropy (NT-Xent) contrastive loss introduced in SimCLR [2]. The objective encourages representations of positive pairs (different augmentations of the same image) to be similar while pushing apart representations of different images.

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\mathrm{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\mathrm{sim}(z_i, z_k)/\tau)} \tag{1}$$

where $\mathrm{sim}(u, v)$ denotes cosine similarity between vectors u and v, τ is a temperature parameter, and N is the batch size.

This learning objective allows the network to learn meaningful representations without using explicit labels.

C. Supervised Fine-Tuning

After contrastive pretraining, the backbone network is fine-tuned for the downstream task of multi-class chest X-ray pathology classification. A linear classification head is added on top of the feature representation produced by the backbone network.

Medical imaging datasets typically exhibit strong class imbalance, where certain diseases occur much less frequently than others. Fig. 3 shows the class distribution of the dataset used in this work. The majority class (*No Finding*) dominates the dataset, while several pathologies such as Pneumomediastinum and Hernia contain only a very small number of samples. This results in an imbalance ratio exceeding several thousand between the most frequent and least frequent classes.

Such imbalance can significantly degrade the performance of models trained using standard cross-entropy loss, as the model tends to bias its predictions toward the majority classes and may fail to detect rare but clinically important conditions.

To address this issue, we employ **Focal Loss** [3], which modifies the standard cross-entropy loss by down-weighting easy examples and focusing training on difficult or misclassified samples.

$$\mathcal{L}_{FL} = -(1 - p_t)^{\gamma} \log(p_t) \tag{2}$$

where p_t is the predicted probability of the ground-truth class and γ is the focusing parameter controlling the strength of the modulating factor.

When an example is correctly classified with high confidence ($p_t \to 1$), the term $(1 - p_t)^{\gamma}$ becomes small, reducing its contribution to the loss. Misclassified or difficult examples

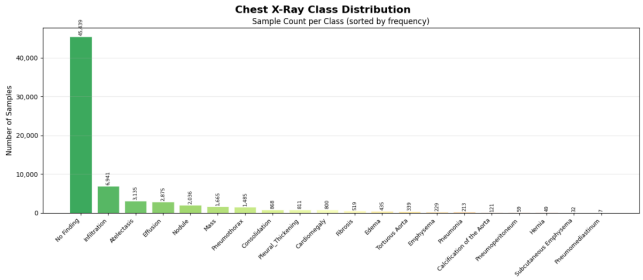

Fig. 3. Class distribution of the chest X-ray dataset. The dataset exhibits severe imbalance, where the "No Finding" class dominates while several pathological classes contain very few samples. This motivates the use of focal loss to improve learning on rare disease classes.

produce larger loss values and receive greater emphasis during training. This helps the model learn better decision boundaries for rare disease classes in highly imbalanced medical datasets.

D. Inference Strategy

During inference, the model outputs class probabilities for each pathology. Prediction performance is further improved using:

- **Threshold Optimization**: Class-specific thresholds are determined using the validation set to balance precision and recall.
- **Test-Time Augmentation**: Predictions are averaged across multiple augmented versions of the input image to improve robustness.

III. Experiments

A. Dataset and Preprocessing

The dataset contains chest X-ray images labeled with multiple thoracic pathologies and exhibits strong class imbalance, with the "No Finding" category dominating while several disease classes have fewer samples.

All images are resized to 384×384 pixels to match EfficientNet-v2 input resolution and normalized using ImageNet statistics. Data augmentation techniques such as random cropping, horizontal flipping, and rotation are applied during training to improve generalization.

The dataset is split into training and validation sets using stratified sampling, with approximately 5% reserved for validation.

B. Training Configuration

Training is performed in two stages. First, contrastive self-supervised pretraining is conducted using the SimCLR framework to learn robust visual representations. In the second stage, the pretrained backbone network is fine-tuned for supervised pathology classification using focal loss.

The key hyperparameters used during training are summarized in Table I.

TABLE I
Training Hyperparameters

Parameter	Value
Backbone Architecture	EfficientNet-v2
Input Resolution	384×384
Pretraining Framework	SimCLR
Projection Head	2-layer MLP
Contrastive Loss	NT-Xent
Temperature (τ)	0.5
Optimizer	AdamW
Learning Rate	3×10^{-4}
Weight Decay	10^{-4}
Batch Size	64
Pretraining Epochs	30
Fine-tuning Epochs	30
Classification Loss	Focal Loss
Focal Loss γ	2.0
Validation Split	5%

C. Training Dynamics

We analyze the behavior of the model during supervised fine-tuning using training loss curves.

After contrastive pretraining, the model is fine-tuned for supervised classification. Fig. 4 shows the training loss during this stage. The smooth decline indicates stable optimization using focal loss.

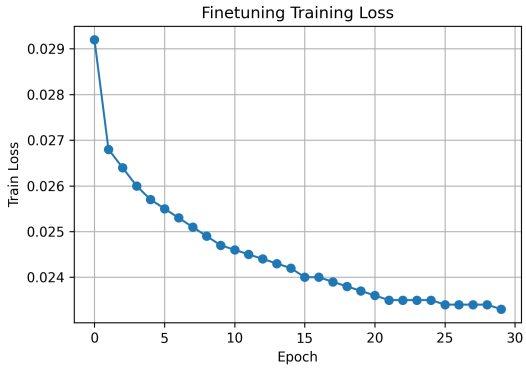

Fig. 4. Training loss during supervised fine-tuning.

Overall, the training curve demonstrates stable convergence and consistent improvement during the fine-tuning stage, validating the effectiveness of the training pipeline.

IV. Results

The proposed training pipeline was evaluated using multiple metrics designed to capture overall classification performance and robustness to class imbalance, which is common in medical imaging datasets.

A. Overall Performance

Table II summarizes the primary validation metrics after fine-tuning. The model achieves a macro AUC-ROC of 0.8287, indicating strong discriminative capability across multiple thoracic pathologies.

TABLE II
OVERALL VALIDATION PERFORMANCE

Metric	Value
Macro AUC-ROC	0.8287
Training Loss	0.5193
Validation Loss	0.7430
Best Threshold Power	0.2
Competition Score	-4.2757

B. Per-Class Diagnostic Performance

To further analyze model behavior across pathologies, the AUC-ROC score was computed for each class. The results are summarized in Table III. Higher scores are observed for conditions with strong radiographic patterns such as edema and pleural effusion.

TABLE III
PER-CLASS AUC-ROC PERFORMANCE

Pathology	AUC	Pathology	AUC
Atelectasis	0.82	Lung Nodule	0.80
Cardiomegaly	0.91	Pleural Thickening	0.78
Consolidation	0.76	Pneumonia	0.60
Edema	0.92	Pneumothorax	0.86
Pleural Effusion	0.90	Lung Opacity	0.95
Emphysema	0.80	Subcutaneous Emphysema	0.94
Fibrosis	0.85	Tortuous Aorta	0.92
Hernia	0.83	Calcification	0.79
Infiltration	0.70	No Finding	0.71
Lung Mass	0.85		

C. Threshold Optimization

Due to class imbalance, classification thresholds significantly influence prediction behavior. A threshold power analysis was performed to study the trade-off between precision, recall, and F1-score.

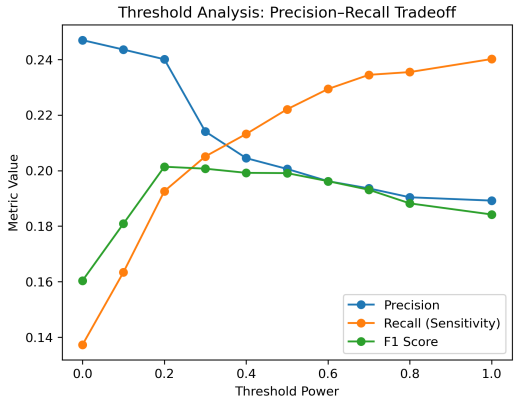

Fig. 5. Precision, recall, and F1-score across different threshold power values.

Lower threshold values increase recall but reduce precision, while higher thresholds produce the opposite effect.

The competition evaluation metric across thresholds is shown in Fig. 6. The best score is obtained at threshold power 0.2, suggesting a favorable balance between sensitivity and specificity.

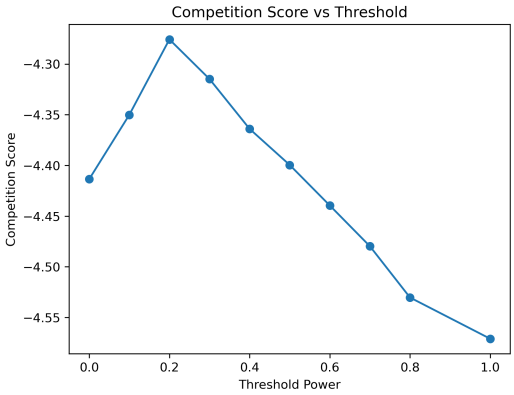

Fig. 6. Competition evaluation score across threshold settings.

D. Discussion

The results demonstrate that contrastive pretraining improves representation learning for chest X-ray images, leading to improved downstream classification performance. Pathologies with distinctive imaging characteristics, such as edema and pleural effusion, achieve higher AUC scores, while visually subtle conditions such as pneumonia remain more challenging.

Threshold optimization further improves clinical usability by adjusting decision boundaries to favor higher sensitivity. This is particularly important in medical settings where minimizing false negatives is critical. Test-time augmentation also contributes to more stable predictions by reducing variance in model outputs.

Overall, the proposed pipeline demonstrates strong performance across multiple thoracic disease categories and highlights the effectiveness of combining contrastive learning with focal loss for imbalanced medical datasets.

V. CONCLUSION

In this assignment, we implemented a deep learning pipeline for multi-class chest X-ray pathology classification. The method combines contrastive self-supervised learning with supervised fine-tuning to learn useful representations from medical images. Contrastive pretraining helps the model learn better features before the final classification stage.

We also addressed challenges such as class imbalance by using focal loss during training. In addition, techniques such as threshold optimization and test-time augmentation were used to improve prediction reliability and overall model performance.

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